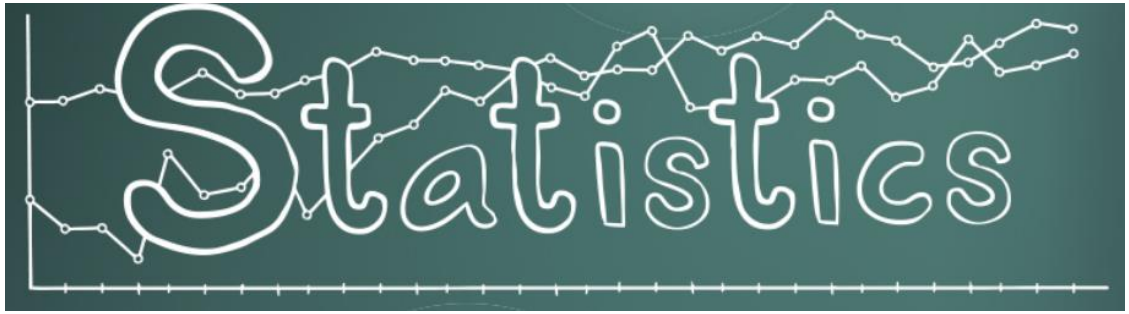



# Mathematical and Statistical Techniques

## Master in Astrophysics, Particle Physics and Cosmology

**Barcelona, autumn 2024**

**X. Luri, L. Garrido**





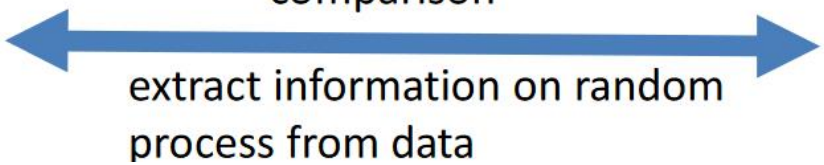
UNIVERSITAT DE BARCELONA

ICCUB
Institut de Ciències del Cosmos

# Mathematical and Statistical Techniques

**Nature**

measurements: $\theta$ +/- $\Delta\theta$




**Theory (*)**

# Mathematical and Statistical Techniques

**Nature**

measurements: $\theta$ +/- $\Delta\theta$

comparison

extract information on random process from data

**Theory (*)**

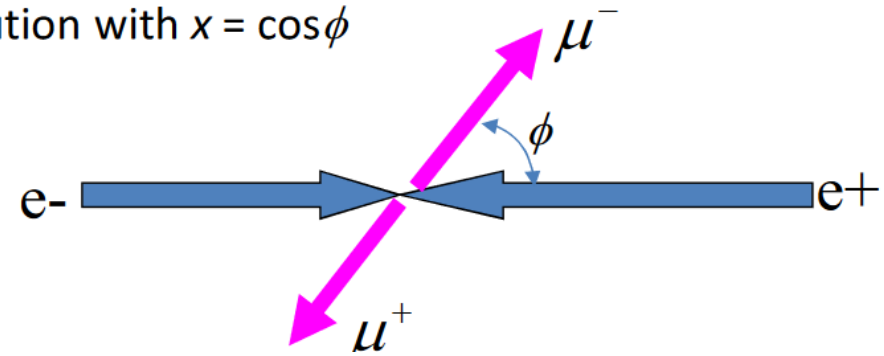
*(*)*

## *Part I: Probability and statistics*

**Theory model f( x ; $\theta$ ).** **f** will normally represent the expected number of observations as a function of some measurable quantities **x** and some variables $\theta$ of the model that must be determined experimentally

**Example:** consider a scattering angle distribution with $x = \cos\phi$

$$f(x;\alpha,\beta) = \frac{1+\alpha x+\beta x^2}{2+2\beta/3}$$



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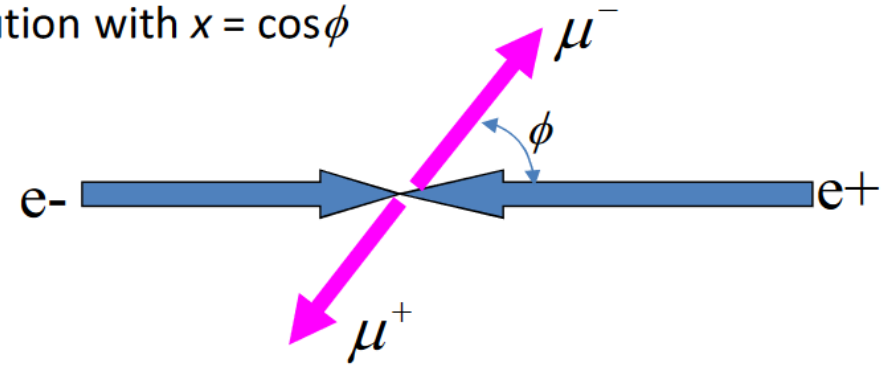
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## *Part II: data analysis*

**NO theory model ->** data analysis techniques: clustering, correlations,...



ICCUB Institute of Cosmos Sciences EXCELENCIA MARÍA DE MAEZTU

# *Part one: fundamentals of probability theory and statistics*

- General review of probability theory
- Monte-Carlo
- Statistical Inference and significance test

Assignments: 1 set of problems + 2 exercises to be simulated in a computer (you can use your preferred language, only a random number generator is needed)

# *Part two: Multivariate Analysis and statistical treatment techniques*

Introduction to multivariate analysis, statistical treatment techniques: concepts and main techniques, hands-on work using specialized software. Data Mining concepts.

- Data analysis and representation
- PCA
- Clustering
- Discriminant analysis
- Neural Networks
- Estimation of the PDF
- Introduction to Data Mining (+Cloud Computing)

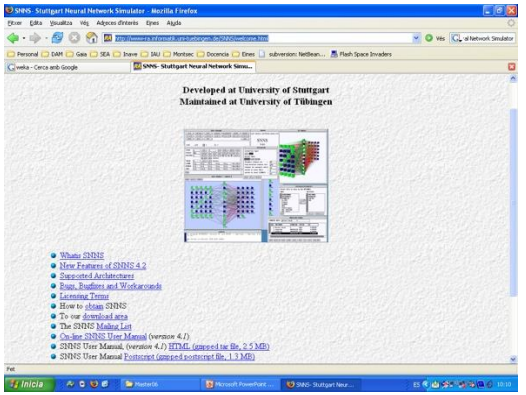
3 data análisis assignments

# Part two: tools

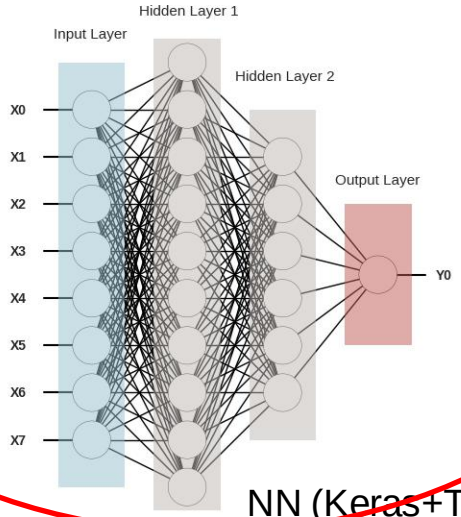
## Weka




## JNNS




## Python

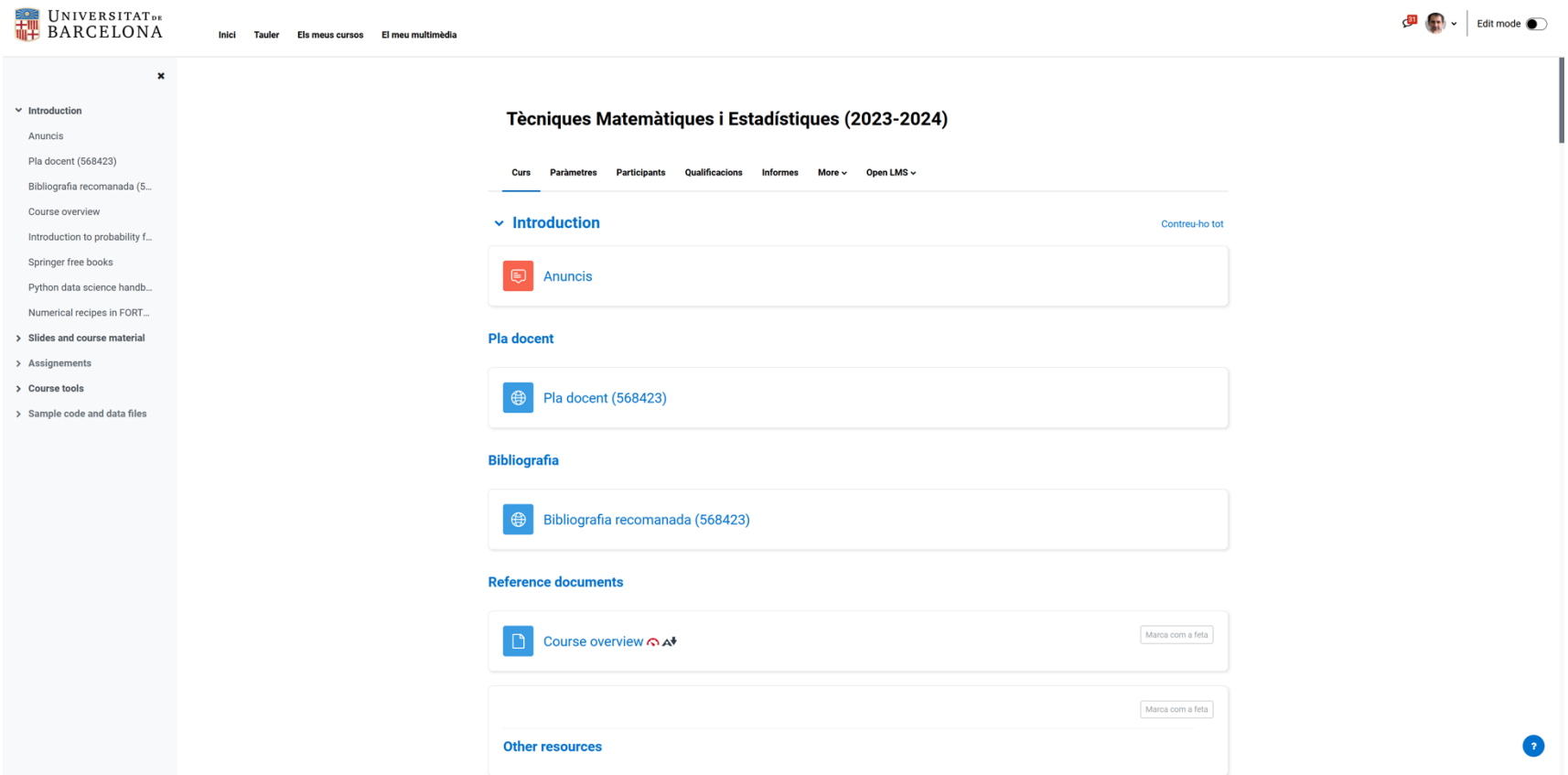



NN (Keras+Tensorflow)

(But you can use any Language*)

# Course Material: Campus virtual

http://campusvirtual.ub.edu



# *Calendar and evaluation*

**6 credits**

Part 1: weeks 1-6
Part 2: weeks 7-12

There will be no final exam for this course.

The average of the 6 problem assignments that will be given during this course, will compose the final global grading.